

Patient Monitoring System (PMS)

Part 1: Appendix A: Functional Requirements

1 Actors

An overview of the actors in the use cases is given in Figure 1. These actors are:

2 Use case diagram

An overview of the use cases is given in Figure 2.

3 Textual use cases

In this section, we model the required functionality of the Patient Monitoring Service (PMS) in the form of *use cases*.

3.1 UC1: log in

- **Name:** log in
- **Primary actor:** User
- **Interested parties:**
 - PMS: wants to authenticate its users.
- **Preconditions:** The User is registered into the system and has credentials to prove their identity.
- **Postconditions:** The User is authenticated to the PMS.
- **Main scenario:**
 1. The User indicates the intention to authenticate to the PMS.
 2. The PMS asks to provide the necessary credentials (e.g., username and password or API token).
 3. The User provides the appropriate credentials.
 4. The PMS verifies the provided credentials and authenticates the User.
- **Alternative scenarios:**
 - 4b. The provided credentials were incorrect, resume at step 2.

3.2 UC2: log off

- **Name:** log off
- **Primary actor:** User
- **Interested parties:**
 - PMS: wants to authenticate its users.

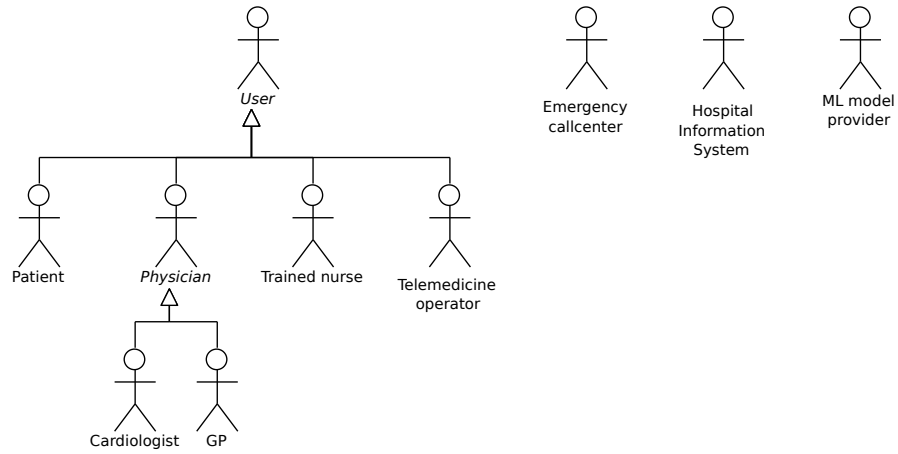


Figure 1: Overview of the actors in the use cases

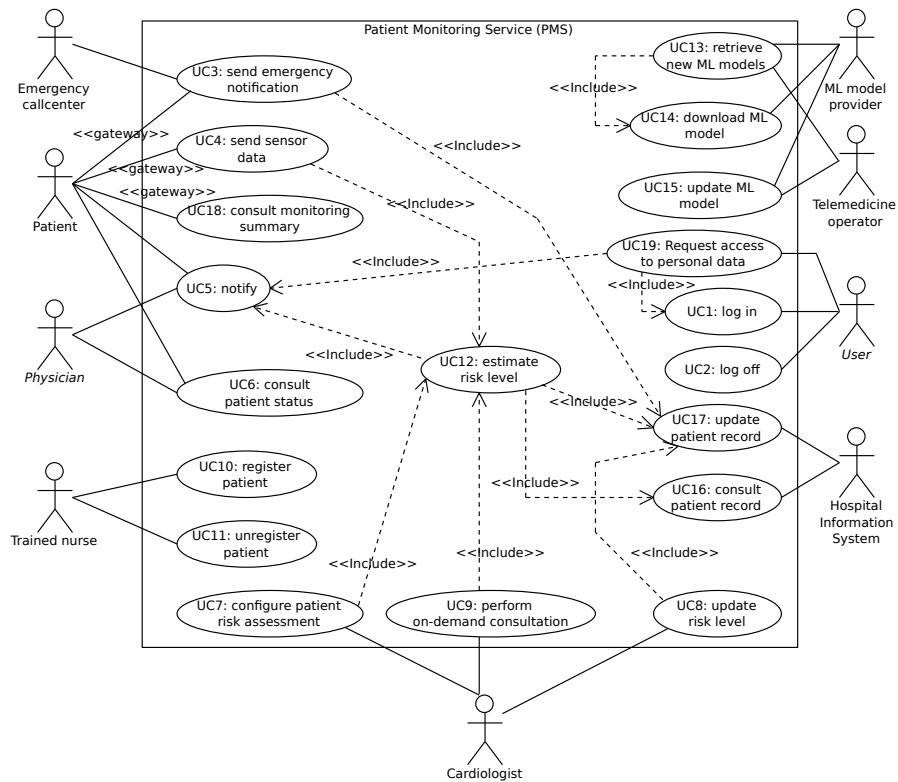


Figure 2: Overview of the use cases.

- **Preconditions:** The User is authenticated to the PMS (cf. *UC1*: log in).
- **Postconditions:** The User has logged off.
- **Main scenario:**
 1. The User indicates they want to log off from the PMS.
 2. The PMS logs them off.
- **Alternative scenarios:** None

3.3 *UC3*: send emergency notification

- **Name:** send emergency notification
- **Primary actor:** Patient (via patient gateway)
- **Secondary actor:** Emergency call center
- **Interested parties:**
 - *Cardiologist*: has diagnosed the patient, decided upon treatment, and enrolled them in the PMS.
 - *Physician*: is involved in the Patient’s treatment.
 - *Emergency call center*: responsible for dealing with emergency situations (e.g., sending an ambulance to the patient).
 - *Patient*: wants to be treated in time when an emergency situation arises.
- **Preconditions:** The patient is registered with the PMS (cf. *UC10*: register patient).
- **Postconditions:** The PMS has received the emergency notification, verified it, and informed the Emergency call center by means of (an updated) emergency notification.
- **Main scenario:**
 1. The patient gateway receives the sensor data from the wearable unit, identifies a potential emergency situation, prepares it (packages the sensor data as an *emergency notification*). There is no need to authenticate explicitly: the patient gateway itself provides a proof of the Patient’s identity to the PMS, for example an API token that uniquely identifies the patient to the system. The patient gateway sends the prepared package to the PMS, together with proof of the Patient’s identity.
 2. The PMS receives the emergency notification and applies a dedicated (yet fast) emergency estimation model for confirming the emergency.
 3. The PMS confirms the emergency, sends an emergency notification with all relevant data to the Emergency call center, and updates the patient record with the relevant information (e.g., type of emergency, date, etc.) (**Include:** *UC17*: update patient record).
- **Alternative scenarios:**
 - 3b. After verifying the emergency notification, the PMS decides there is no need to issue an emergency notification but marks this event in the patient record (**Include:** *UC17*: update patient record).

3.4 *UC4*: send sensor data

- **Name:** send sensor data
- **Primary actor:** Patient (via patient gateway)

- **Interested parties:**
 - *Cardiologist*: has diagnosed the patient, decided upon treatment, and enrolled them in the PMS.
 - *Physician*: is involved in the Patient’s treatment.
 - *Patient*: wants the system to keep an eye on the evolution of their CVD.
- **Preconditions:** The patient is registered with the PMS (cf. *UC10*: register patient).
- **Postconditions:** The PMS has registered the sensor data (readings) and processes them.
- **Main scenario:**
 1. The patient gateway has received the sensor data from the wearable unit, prepared them (packaging). There is no need to authenticate explicitly: the patient gateway itself provides a proof of the Patient’s identity to the PMS. The patient gateway sends the packaged sensor data to the PMS (according to the configured transmission rate), together with proof of the Patient’s identify.
 2. The PMS receives the sensor data, stores it and schedules it for processing (**Include**: *UC12*: estimate risk level).
- **Alternative scenarios:** None

3.5 *UC5*: notify registered parties

- **Name:** notify registered parties
- **Primary actor:** PMS
- **Interested parties:**
 - *Cardiologist*: has diagnosed the patient, decided upon treatment, and enrolled them in the PMS.
 - *Physician*: is involved in the Patient’s treatment
 - *Patient*: wants to be up-to-date about their status.
- **Preconditions:** The PMS has previously received sensor data (cf. *UC4*: send sensor data), processed it (*UC12*: estimate risk level), and determined that notifications should be sent.
- **Postconditions:** The registered parties will have received a notification about the Patient’s status.
- **Main scenario:**
 1. The PMS determines and looks up the registered parties interested in notifications (Physicians, Cardiologist, and the Patient himself).
 2. The PMS prepares a notification specifically for the registered party (e.g., the cardiologist receives a detailed medical notification, while the Patient only gets a summary), and sends out these notifications.
 3. The registered party receives the notification.
- **Alternative scenarios:** None

3.6 *UC6*: consult patient status

- **Name:** consult patient status
- **Primary actor:** Physician or Patient
- **Interested parties:**

- *Cardiologist*: has diagnosed the patient, decided upon treatment, and enrolled them in the PMS.
- *Physician*: is involved in the Patient’s treatment and wants to be up-to-date.
- *Patient*: wants to be up-to-date about their status.
- **Preconditions:** The primary actor is authenticated (cf. *UC1*: log in).
- **Postconditions:**
 - The primary actor has consulted a patient’s status.
 - The PMS has logged this event.
- **Main scenario:**
 1. The primary actor indicates they want to consult the current status of a patient.
 2. The PMS looks up the information, and presents it to the primary actor, tailored specifically to the expertise level of the primary actor (e.g., the Cardiologist receives a detailed medical information, while the Patient only gets a summary).
 3. The PMS logs this event.
- **Alternative scenarios:** None

3.7 *UC7*: configure patient risk assessment

- **Name:** configure patient risk assessment
- **Primary actor:** Cardiologist
- **Interested parties:**
 - *Cardiologist*: has diagnosed the patient, decided upon treatment, and enrolled them in the PMS.
 - *Patient*: is monitored by the PMS to estimate their risk level.
- **Preconditions:**
 - The Cardiologist is authenticated (cf. *UC1*: log in).
 - The Cardiologist is watching a patient’s status (cf. *UC6*: consult patient status).
- **Postconditions:**
 - The PMS has registered the configuration and will process the sensor data from the patient accordingly.
 - The PMS has logged this event.
- **Main scenario:**
 1. The Cardiologist requests to configure the patient’s risk assessment.
 2. The PMS presents the Cardiologist with an overview of the configurable options with their current value, if any. This includes, for example, which measurements should be monitored for the patient, e.g., heart rate and blood pressure, any relevant thresholds, e.g., typical rest heart rate, and underlying conditions that should be considered such as hypertension.
 3. The Cardiologist changes one or more configuration options and confirms the (re)configuration.
 4. The PMS stores the new configuration and logs this event.
 5. The PMS schedules a recalculation of the patient’s risk level with the updated configuration (**Include:** *UC12*: estimate risk level).
- **Alternative scenarios:** None

3.8 UC8: update risk level

- **Name:** update risk level
- **Primary actor:** Cardiologist
- **Interested parties:**
 - *Cardiologist*: has diagnosed the patient, decided upon treatment, and enrolled them in the PMS.
 - *Patient*: is monitored by the PMS to estimate their risk level.
- **Preconditions:**
 - The Cardiologist is authenticated (cf. UC1: log in).
 - The Cardiologist may have received a notification from the PMS providing risk estimates for a patient (cf. UC5: notify registered parties).
 - The Cardiologist is watching a patient's status (cf. UC6: consult patient status).
- **Postconditions:**
 - The PMS has adapted the risk level of the patient.
 - The PMS has logged this event.
- **Main scenario:**
 1. The Cardiologist indicates the need to change the risk level of a patient.
 2. The PMS presents the Cardiologist with the option to change risk level to low, medium, or high.
 3. The Cardiologist indicates which risk level the patient's risk level should be changed to.
 4. The PMS changes the risk level of the Patient and updates the patient record (**Include:** UC17: update patient record).
- **Alternative scenarios:**
 - 1b. The Cardiologist indicates they wants to accept the risk level estimation given by the PMS in a notification. Forward to step 4.

3.9 UC9: perform on-demand consultation

- **Name:** perform on-demand consultation
- **Primary actor:** Cardiologist
- **Interested parties:**
 - *Cardiologist*: has diagnosed the patient, decided upon treatment, and enrolled them in the PMS.
 - *Patient*: is monitored by the PMS to estimate their risk level.
- **Preconditions:**
 - The Cardiologist is authenticated (cf. UC1: log in).
 - The Cardiologist is watching a patient's status (cf. UC6: consult patient status).
- **Postconditions:** The Cardiologist has performed an on-demand consultation.
- **Main scenario:**
 1. The Cardiologist requests to receive the current data for a patient from the PMS.
 2. The PMS issues a request to the patient's gateway for current sensor data.
 3. The Patient gateway responds by sending the current sensor data.

4. The PMS schedules the processing of the sensor data (**Include:** *UC12*: estimate risk level) and presents the results of the on-demand consultation to the Cardiologist.

- **Alternative scenarios:** None

3.10 *UC10*: register patient

- **Name:** register patient
- **Primary actor:** Trained nurse
- **Interested parties:**
 - *Cardiologist*: has diagnosed the patient and want to register them in the PMS.
 - *Patient*: is suffering from CVD and wants to be monitored by the PMS.
 - *Trained nurse*: executes the registration process.
- **Preconditions:**
 - The Cardiologist has approved the patient’s registration.
 - The Trained nurse is authenticated (cf. *UC1*: log in).
- **Postconditions:**
 - The patient is registered in the PMS.
 - The PMS has logged this event.
- **Main scenario:**
 1. The Trained nurse indicates to the PMS a registration process should be started for a patient (e.g., by selecting the appropriate patient record).
 2. The PMS asks to select two devices and asks her to pair them.
 3. The Trained nurse selects a wearable unit and a patient gateway for the patient, pairs both devices. The Trained Nurse also links the patient gateway to the identity of the patient (e.g., serial number of the gateway, or unique identifier) for authentication purposes.
 4. The PMS initializes the system, performs diagnostic tests, requests first sensor data from the gateway, and shows these to the Trained nurse.
 5. The Trained nurse double-checks the sensor data and, if necessary, calibrates the sensors.
 6. The PMS allows the nurse to set authentication credentials (e.g., username and password) for the patient.
 7. The Trained nurse enters the authentication credentials.
 8. The PMS registers the patient.

- **Alternative scenarios:** None

3.11 *UC11*: deregister patient

- **Name:** deregister patient
- **Primary actor:** Trained nurse
- **Interested parties:**
 - *Cardiologist*: has diagnosed the patient and wants them to stop using the PMS.
 - *Patient*: has improved sufficiently and can be deregistered from the PMS.
 - *Trained nurse*: executes the deregistration process.

- **Preconditions:**
 - The Patient is registered in the PMS (cf. *UC10*: register patient).
 - The Trained nurse is authenticated (cf. *UC1*: log in).
 - The Cardiologist has approved the Patient’s deregistration.
- **Postconditions:**
 - The Patient is deregistered from the PMS.
 - The PMS has logged this event.
- **Main scenario:**
 1. The Trained nurse indicates to the PMS a patient should be deregistered (e.g., by selecting the appropriate patient record).
 2. The PMS deregisters the wearable unit and patient gateway, deregisters the patient (stops the monitoring), and logs this event.
- **Alternative scenarios:** None

3.12 *UC12*: estimate risk level

- **Name:** estimate risk
- **Primary actor:** PMS
- **Interested parties:**
 - *Cardiologist*: wants to keep track of a patient’s status, and if necessary, adjust treatment.
 - *Physician*: wants to keep track of a patient’s status.
 - *Patient*: is suffering CVD and wants to be monitored by the PMS.
- **Preconditions:**
 - The PMS has received new sensor data (cf. *UC4*: send sensor data) for a patient.
 - The cardiologist has configured the patient’s risk assessment (cf. *UC7*: configure patient risk assessment).
- **Postconditions:** The PMS updated the patient’s status, and if necessary, issued a notification to the interested parties.
- **Main scenario:**
 1. The PMS looks up the patient’s monitoring history, current status, and patient record (medication, etc.) (**Include:** *UC16*: consult patient record), and risk assessment configuration.
 2. The PMS applies all models in the risk assessment to the newly arrived information and the already-known data.
 3. The PMS determines that the patient’s estimated risk level should change based on the combined result from the models in the risk assessment.
 4. The PMS determines the interested parties (e.g., Cardiologist, Physicians, and Patient) and issues the necessary notifications (**Include:** *UC5*: notify registered parties).
 5. The PMS determines that the newly arrived sensor data and the results of the risk assessment should propagate to the patient record and sends an appropriate update (**Include:** *UC17*: update patient record).
- **Alternative scenarios:**
 - 3b. The PMS determines that the patient’s estimated risk level should **not** change based on the combined result from the models in the risk assessment. Skip to step 5.

- 5b. The PMS determines that the newly arrived sensor data and the results of risk estimation should **not** propagate to the patient record. Use case ends.

3.13 *UC13*: retrieve new ML models

- **Name:** retrieve new ML models
- **Primary actor:** Telemedicine operator
- **Secondary actor:** ML model provider
- **Interested parties:**
 - *Physician*: wants to ensure the most appropriate medical care is given to patients.
- **Preconditions:** The Telemedicine operator is authenticated to the system (cf. *UC1*: log in).
- **Postconditions:** The PMS has retrieved and stored newly available ML models.
- **Main scenario:**
 1. The Telemedicine operator indicates to the system to check for newly available ML models.
 2. The PMS queries the model provider for a list of available ML models.
 3. The ML model provider replies with a list of all available ML models, including metadata including an identifier and version number.
 4. The PMS determines that new ML models are available and, for each such model, the PMS provides the Telemedicine operator with an option to download the new ML model from the ML model provider (**Include:** *UC14*: download ML model)
- **Alternative scenarios:**
 - 4b. The PMS determines no new ML models are available and informs the Telemedicine operator of this. Use case ends.

3.14 *UC14*: download ML model

- **Name:** download ML model
- **Primary actor:** PMS
- **Secondary actor:** ML model provider
- **Interested parties:**
 - *Physician*: wants to ensure the most appropriate medical care is given to patients.
- **Preconditions:** None
- **Postconditions:** An ML model was retrieved from the ML model provider and installed into the system.
- **Main scenario:**
 1. The PMS queries the ML model provider for a specific ML model by providing by model identifier.
 2. The ML model provider replies with the corresponding ML model.
 3. The PMS adds the received model to its repository of installed ML models.
- **Alternative scenarios:** None

3.15 *UC15*: update ML model

- **Name:** Update ML model
- **Primary actor:** Telemedicine operator
- **Secondary actor:** ML model provider
- **Interested parties:**
 - *Physician*: wants to ensure the most appropriate medical care is given to patients.
- **Preconditions:** The Telemedicine operator is authenticated to the system (cf. *UC1*: log in).
- **Postconditions:** The PMS has updated an available ML model.
- **Main scenario:**
 1. The Telemedicine operator requests a list of ML models available in the system.
 2. The PMS presents the Telemedicine operator with the list of available ML models.
 3. The Telemedicine operator selects a specific model to inspect in more detail.
 4. The PMS retrieves the details of the specified ML model and shows this to the Telemedicine operator.
 5. The Telemedicine operator indicates that the model should be updated.
 6. The PMS queries the ML model provider to retrieve all updates available for the specified model.
 7. The ML model provider replies with the requested updates.
 8. The system applies the received updates to the installed ML model.
 9. The PMS informs the Telemedicine operator that the ML model was updated.
- **Alternative scenarios:**
 - 2b. No ML models are currently installed in the system, the Telemedicine operator is informed of this. Use case ends.
 - 7b. The model provider replies that no updates are available for the selected ML model. The system informs the Telemedicine operator that the installed ML model is already the most recent version. Use case ends.

3.16 *UC16*: consult patient record

- **Name:** consult patient record
- **Primary actor:** PMS
- **Secondary actor:** Hospital Information System (HIS)
- **Interested parties:**
 - PMS: wants to keep track of a patient's status, also the information external to the PMS itself
 - HIS: provides access to the patient record.
 - *Physician*: wants to have access to up-to-date information about the patient.
- **Preconditions:** The patient is registered with the PMS (cf. *UC10*: register patient).

- **Postconditions:** The PMS has received the patient record from the HIS.
- **Main scenario:**
 1. The PMS requests a look-up of the patient record from the HIS.
 2. The HIS service performs the look-up and provides the patient record to the PMS.
- **Alternative scenarios:** None

3.17 *UC17*: update patient record

- **Name:** update patient record
- **Primary actor:** PMS
- **Secondary actor:** Hospital Information System (HIS)
- **Interested parties:**
 - PMS: wants to update the patient record at the hospital with relevant information about the patient's health (e.g., certain events, summarized data, etc.).
 - HIS: provides access to the patient record.
 - *Physician*: wants to keep the patient (medical) record up-to-date.
- **Preconditions:** The PMS is monitoring a patient of the hospital.
- **Postconditions:**
 - The PMS has provided information for the patient record in the HIS.
- **Main scenario:**
 1. The PMS sends an update to the patient record in the HIS.
 2. The HIS accepts the data from PMS and processes it further.
- **Alternative scenarios:** None

3.18 *UC18*: consult monitoring summary

- **Name:** consult monitoring summary
- **Primary actor:** Patient
- **Interested parties:** None
- **Preconditions:** The User is authenticated to the PMS (cf. *UC1*: log in).
- **Postconditions:**
 - The PMS has provided the patient with a summary of their information in the system.
- **Main scenario:**
 1. The Patient indicates, via their gateway, that they want a summary of their monitoring information.
 2. The PMS retrieves the patient's information and compiles a summary suited for a layperson and sends this to the patient gateway. This summary can contain, for example, a graph showing the risk level changes over time.
 3. The patient gateway shows the received summary to the patient, e.g., as tables and/or graphs.
- **Alternative scenarios:** None

3.19 UC19: Request access to personal data

- **Name:** request access to personal data
- **Primary actor:** User
- **Interested parties:** None
- **Preconditions:** None
- **Postconditions:**
 - The primary actor is informed on whether the PMS has processed their personal data at some point.
 - The primary actor is provided with a personal data archive that can be downloaded.
- **Main scenario:**
 1. The primary actor indicates to the system they want to perform a right to access request.
 2. The primary actor authenticates to the PMS (**Include:** UC1: log in).
 3. The PMS determines that it processes, or has processed, personal data concerning the primary actor.
 4. The system informs the primary actor that it processes personal data concerning them (**Include:** UC5: notify registered parties).
 5. The PMS gathers all relevant personal data into a personal data archive. This includes at least the following data, along with information on the purpose of processing this data:
 - profile information, including personal data kept on the patient Gateway;
 - all risk estimations;
 - all sensor data readings;
 - data used from the patient’s electronic health record; and
 - patient-specific configuration for the risk estimations, for example a clinical model tailored to the patient in question.
 6. The system notifies the the primary actor their personal data archive is available for download (**Include:** UC5: notify registered parties).
- **Alternative scenarios:**
 - 2b. The primary actor provides an email address for identification, e.g., because they are currently not registered with the PMS.
 - 2b.1 The PMS sends an email containing a token to verify that the requesting user owns the provided email address.
 - 2b.2 The primary actor provides the emailed token to the PMS to confirm they have access to the provided email address. Continue with step 3.
 - 3b. The PMS determines that it has not processed personal data concerning the primary actor and informs the primary actor of this. The use case ends.
 - 6b. If the primary actor provided an email address for identification, the PMS sends an email containing a link that can be used to download the archive.